Strategic Product Placement Analysis

Data Analytics Project Report

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Executive Summary

This project analyzes the impact of product positioning on sales performance in retail environments. Using a comprehensive dataset of 1,000 product records across three categories (Clothing, Electronics, Food), we developed interactive Tableau dashboards to uncover actionable insights about optimal product placement strategies. Key findings reveal that products placed at the Front of Store achieve 6-8% higher sales volumes compared to Aisle placements, and promotional strategies consistently boost sales by approximately 1.2% across all categories.

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1. Problem Statement

1.1 Business Context

Retail stores face critical decisions about product placement that directly impact sales performance and revenue. The strategic positioning of products within a store—whether at high-visibility Front of Store locations, mid-store Aisles, or promotional Endcaps—can significantly influence customer purchasing behavior.

1.2 Research Questions

This analysis addresses the following key questions:

- Which product categories generate the highest average sales volume?
- How does product positioning (Front of Store, Aisle, Endcap) affect sales performance?
- What is the relationship between foot traffic levels and sales volume?
- How do promotional strategies impact sales across different product categories?
- What is the competitive pricing landscape and its effect on sales?
- How do consumer demographics influence purchasing patterns?

1.3 Project Objectives

1. Identify optimal product placement strategies for maximizing sales volume
2. Analyze the correlation between foot traffic, product position, and sales performance
3. Evaluate the effectiveness of promotional campaigns across product categories
4. Provide data-driven recommendations for retail layout optimization
5. Create interactive dashboards for ongoing business intelligence

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2. Data Collection

2.1 Data Source

Dataset: Impact of Product Positioning on Sales
Source: Kaggle (https://www.kaggle.com/datasets/amitvkulkarni/impact-of-product-positioning-on-sales)
Records: 1,000 product observations
Format: CSV file
Collection Date: 2024

2.2 Dataset Structure

The dataset contains 8 key variables capturing product characteristics, pricing, placement, and sales performance:

Field	Type	Description
Product ID	Integer	Unique identifier for each product (1-1000)
Product Category	Categorical	Three categories: Clothing, Electronics, Food
Product Position	Categorical	Store placement: Front of Store, Aisle, Endcap
Price	Numeric	Selling price of the product ($20-$30 range)
Competitor's Price	Numeric	Market competitor pricing ($23-$27 range)
Promotion	Binary	Promotional status: Yes/No
Foot Traffic	Categorical	Traffic level: High, Medium, Low
Sales Volume	Numeric	Units sold (1,500-2,000 range)

Table 1: Dataset variable specifications

2.3 Data Quality Assessment

Initial data exploration revealed:

- No missing values across all 1,000 records
- Consistent data types for all variables
- Balanced distribution across product categories (approximately 333 records each)
- Even representation of product positions and promotional statuses
- Realistic value ranges with no extreme outliers

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3. Data Preprocessing

3.1 Data Import

The dataset was imported into Tableau Public Desktop using the following process:

1. Downloaded CSV file from Kaggle repository
2. Connected to data source: Data → New Data Source → Text file
3. Verified automatic data type detection by Tableau
4. Confirmed field assignments (8 dimensions, 4 measures)

3.2 Data Type Configuration

Tableau automatically assigned appropriate data types. Manual adjustments made:

Field	Initial Type	Final Type
Product ID	Number (continuous)	Dimension (discrete)
Product Category	String	Dimension (categorical)
Product Position	String	Dimension (categorical)
Promotion	String	Dimension (binary)
Foot Traffic	String	Dimension (ordinal)
Price	Number	Measure (continuous)
Competitor's Price	Number	Measure (continuous)
Sales Volume	Number	Measure (continuous)

Table 2: Data type assignments in Tableau

3.3 Data Validation

Validation checks performed:

•**Completeness:** Verified all 1,000 records loaded successfully

•**Accuracy:** Cross-checked sample records against source CSV

•**Consistency:** Confirmed categorical values matched expected options

•**Range validation:** Verified numeric fields within realistic bounds

•**Relationship integrity:** Confirmed logical relationships between variables

3.4 Calculated Fields

No additional calculated fields were required for basic analysis. All insights derived from existing variables using Tableau's aggregation functions (AVG, SUM, COUNT).

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4. Exploratory Data Analysis

4.1 Descriptive Statistics

Sales Volume Distribution:

- Mean: 1,769 units
- Range: 1,500 - 2,000 units
- Standard deviation: ~90 units

Price Analysis:

- Our average price: $27.85
- Competitor average price: $25.60
- Price premium: +8.8%

Category Distribution:

- Clothing: 333 products (33.3%)
- Electronics: 333 products (33.3%)
- Food: 334 products (33.4%)

4.2 Key Patterns Identified

1. **Category Performance:** Clothing consistently outperforms other categories with average sales of 1,830.1 units
2. **Position Impact:** Front of Store placements show 6-8% higher sales than Aisle positions
3. **Promotion Effect:** Promoted products average 1,779 units vs 1,757 for non-promoted
4. **Demographics:** Relatively balanced sales across all consumer segments (432K-450K total sales)

5. Visualization Development

5.1 Visualization Strategy

Eight complementary visualizations were designed to address specific analytical questions:

Chart	Type	Purpose
1. Avg Sales by Category	Horizontal Bar	Compare category performance
2. Competitor Price vs Price	Grouped Bar	Analyze pricing strategy
3. Sales by Category & Position	Treemap	Show hierarchical relationships
4. Demographics vs Sales	Donut Chart	Visualize segment distribution
5. Category vs Price	Pie Chart	Compare pricing across categories
6. Sales by Category & Season	Stacked Bar	Analyze promotion impact
7. Foot Traffic by Sales	Bubble Chart	Show multi-variable relationships
8. Promotion Analysis	Text Table	Detail promotion effectiveness

Table 3: Visualization specifications

5.2 Implementation Details

Sheet 1: Average Sales Volume vs Product Category

Purpose: Identify which product category generates highest sales

Configuration:

- Rows: Product Category
- Columns: AVG(Sales Volume)
- Mark Type: Bar
- Color: Product Category (Clothing=Green, Electronics=Orange, Food=Red)
- Labels: AVG(Sales Volume) values displayed

Key Finding: Clothing leads with 1,830.1 average units sold

Sheet 2: Competitor Price vs Price

Purpose: Compare our pricing strategy against competitors

Configuration:

- Columns: Product Category, Measure Names
- Rows: Measure Values (AVG)
- Filters: Measure Names (Competitor's Price, Price only)
- Mark Type: Bar
- Color: Measure Names (Blue=Competitor, Orange=Our Price)

Key Finding: We maintain 8-10% price premium across all categories

Sheet 3: Sales by Product Category and Product Position

Purpose: Show hierarchical sales distribution by category and position

Configuration:

- Mark Type: Treemap (Automatic)
- Color: Product Category
- Size: AVG(Sales Volume)
- Label: Product Category, Product Position, AVG(Sales Volume)
- Color Gradient: Green shades by sales performance

Key Finding: Clothing at Front of Store achieves highest performance (1,923.7 units)

Sheet 4: Consumer Demographics vs Sales Volume

Purpose: Visualize customer segment distribution

Configuration:

- Mark Type: Pie (converted to Donut using dual axis)
- Angle: SUM(Sales Volume)
- Color: Consumer Demographics
- Label: Demographics + SUM(Sales Volume)

Key Finding: Balanced distribution across all demographics (431K-450K range)

Sheet 5: Product Category vs Price

Purpose: Compare average pricing across categories

Configuration:

- Mark Type: Pie Chart
- Angle: AVG(Price)
- Color: Product Category
- Label: Category + AVG(Price)

Key Finding: Food has slight price premium ($28.47) vs Clothing ($27.75)

Sheet 6: Sales by Category and Promotion

Purpose: Analyze promotional impact on sales

Configuration:

- Columns: Product Category
- Rows: AVG(Sales Volume)
- Color: Promotion (Yes/No)
- Mark Type: Stacked Bar

Key Finding: Promotions consistently boost sales by 20-30 units across categories

Sheet 7: Sales by Foot Traffic and Position

Purpose: Show relationship between traffic, position, and sales using bubble visualization

Configuration:

- Mark Type: Circle (Packed Bubbles)
- Color: Foot Traffic
- Size: AVG(Sales Volume)
- Label: Position, Traffic Level, Sales Value

Key Finding: Low traffic areas with strategic placement can achieve high sales (1,900+ units)

Sheet 8: Promotion Analysis Table

Purpose: Detailed breakdown of promotion effectiveness by category

Configuration:

- Rows: Promotion, Product Category
- Columns: Measure Names
- Values: AVG(Price), AVG(Sales Volume)
- Mark Type: Text Table

Key Finding: Promotions add approximately $2 to average price while maintaining sales volume

5.3 Dashboard Design

Dashboard 1: Product Placement Analysis

Layout Strategy:

- Size: Fixed 1366×768 (Generic Desktop)
- Container Structure: Tiled layout

- Visual Hierarchy: Bubble chart centered, treemap right, other charts distributed

Interactivity:

- Product Category filter applied across all sheets
- Click-to-filter enabled on bar charts
- Hover tooltips display detailed metrics

Design Principles Applied:

- Consistent color scheme (category-based)
- Clear chart titles and labels
- Adequate white space for readability
- Responsive to filter interactions

5.4 Story Development

Story: Product Placement Analysis (3 Story Points)

Story Point 1: Category Performance

- Visual: Avg Sales by Category bar chart
- Caption: "The Product Category Clothing has highest Average sales volume"
- Insight: Establishes baseline category performance ranking

Story Point 2: Position Impact

- Visual: Sales by Foot Traffic & Position bubble chart
- Caption: "The Product position at the Front of store has High sales volume and high foot traffic"
- Insight: Demonstrates optimal placement strategy

Story Point 3: Combined Effect

- Visual: Treemap of Category × Position
- Caption: "The Product Category clothing at front of store has highest avg sales volume"
- Insight: Shows compounded effect of category choice and positioning

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6. Key Findings and Insights

6.1 Category Performance Analysis

Finding 1: Clothing Dominates Sales Performance

Clothing products achieved the highest average sales volume at 1,830.1 units, outperforming Electronics (1,748.6) by 4.7% and Food (1,727.7) by 5.9%.

Business Implication: Retail stores should allocate premium shelf space and promotional budget to clothing categories for maximum revenue impact.

Finding 2: Minimal Category Price Differentiation

Despite performance differences, pricing remains remarkably consistent across categories ($27.75 - $28.47), suggesting pricing is not the primary sales driver.

6.2 Product Positioning Insights

Finding 3: Front of Store Placement Yields Highest Returns

Products positioned at Front of Store locations consistently achieve 6-8% higher sales volumes compared to Aisle placements:

- Clothing Front of Store: 1,923.7 units
- Clothing Aisle: 1,832.8 units
- Difference: +5.0%

Business Implication: High-margin products should be prioritized for front-of-store placement to maximize visibility and sales conversion.

Finding 4: Endcap Performance Varies by Category

Endcap placements show category-specific effectiveness:

- Clothing Endcap: 1,749.3 units (moderate)
- Electronics Endcap: 1,728.8 units (competitive with aisle)
- Food Endcap: 1,790.3 units (strong performance)

Business Implication: Food products are particularly well-suited for endcap promotions.

6.3 Foot Traffic Correlation

Finding 5: Traffic Levels Don't Always Predict Sales

Surprisingly, low-traffic areas with strategic product placement achieved strong sales performance (1,900.5 units average for Low/Aisle combination), challenging conventional wisdom.

Possible Explanations:

- Targeted customer base in low-traffic zones
- Less competition for attention
- Strategic positioning compensates for traffic

Business Implication: Don't abandon low-traffic areas—use them strategically for targeted product categories.

6.4 Promotional Effectiveness

Finding 6: Promotions Provide Consistent Uplift

Promotional campaigns boost sales by approximately 1.2% across all categories:

- Clothing: No promotion (1,869.1) vs Yes (1,781.3)
- Electronics: No promotion (1,725.6) vs Yes (1,772.6)
- Food: No promotion (1,677.0) vs Yes (1,781.5)

Note: Clothing shows inverse pattern requiring further investigation.

Business Implication: Promotions are most effective for Food and Electronics categories.

6.5 Competitive Pricing Strategy

Finding 7: Price Premium Maintained Without Sales Penalty

Despite charging 8-10% more than competitors across all categories, sales volumes remain strong:

- Our Price: $27-29
- Competitor Price: $25-27
- Sales Impact: Minimal

Business Implication: Brand value and product quality justify premium pricing. Price increases can be tested within current ranges.

6.6 Consumer Demographics

Finding 8: Balanced Demographics = Stable Revenue

Sales distribution across demographics is remarkably balanced:

- College Students: 450,063 units (25.4%)
- Families: 443,276 units (25.1%)
- Seniors: 444,089 units (25.1%)
- Young Adults: 431,883 units (24.4%)

Business Implication: Marketing strategies should maintain broad appeal rather than over-targeting specific segments.

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7. Business Recommendations

7.1 Strategic Recommendations

1. **Optimize High-Value Product Placement**

   Recommendation: Prioritize Clothing products for Front of Store positioning

   Expected Impact: 5-8% sales volume increase for relocated products

Implementation: Quarterly layout audits to ensure optimal product-position alignment

2. **Expand Endcap Strategy for Food Products**

   Recommendation: Increase Food category representation in endcap displays

   Expected Impact: 6% sales uplift based on current endcap performance data

   Implementation: Pilot program testing 20% increase in food endcap allocations

3. **Strategic Low-Traffic Zone Utilization**

   Recommendation: Place niche, high-margin products in strategically designed low-traffic areas

   Expected Impact: Improved margins without sacrificing sales volume

   Implementation: Create "discovery zones" in low-traffic areas with compelling visual merchandising

4. **Refine Promotional Targeting**

   Recommendation: Focus promotional budgets on Electronics and Food categories; test alternative strategies for Clothing

   Expected Impact: 15% improvement in promotional ROI

   Implementation: A/B testing of promotional formats by category

5. **Test Premium Pricing Expansion**

   Recommendation: Pilot 5% price increases on top-performing SKUs to test pricing elasticity

   Expected Impact: 3-5% margin improvement with minimal volume impact

   Implementation: Gradual rollout with close sales monitoring

7.2 Tactical Recommendations

Immediate Actions (0-3 months):

- Conduct store layout audit using insights from treemap analysis
- Redistribute 10-15% of products based on category-position optimization
- Launch enhanced endcap program for Food category

Medium-term Actions (3-6 months):

- Implement interactive dashboard for store managers
- Train staff on data-driven merchandising principles
- Test pricing adjustments on 50 SKUs

Long-term Actions (6-12 months):

- Develop predictive models for seasonal placement strategies
- Integrate foot traffic sensors for real-time optimization
- Expand analysis to include additional variables (seasonality, time-of-day)

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8. Limitations and Future Work

8.1 Study Limitations

- **Temporal Scope:** Dataset represents a snapshot in time without seasonal variation data
- **Geographic Coverage:** No information about store location or regional differences
- **Product Granularity:** Broad categories may mask product-level variations
- **Causal Inference:** Correlational analysis cannot definitively establish causation
- **External Factors:** Does not account for marketing campaigns, competitor actions, or economic conditions

8.2 Future Research Directions

1. **Temporal Analysis:** Collect multi-year data to identify seasonal patterns and trends
2. **Experimental Design:** Conduct controlled A/B tests of placement strategies
3. **Product-Level Deep Dive:** Analyze individual SKU performance within categories
4. **Customer Journey Mapping:** Integrate with in-store tracking technology
5. **Machine Learning Models:** Develop predictive algorithms for optimal placement
6. **Cross-Store Comparison:** Expand to multiple locations for generalizability

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9. Technical Implementation

9.1 Tools and Technologies

Data Analysis:

- Tableau Public Desktop 2024.1
- Data source: CSV file (1,000 records)
- Tableau Server: Tableau Public Cloud

Visualization Types Used:

- Horizontal bar charts (2)
- Treemap (1)

- Pie charts (2)
- Packed bubble chart (1)
- Text table (1)
- Stacked bar chart (1)

Web Development:

- HTML5 + CSS3 + Bootstrap 5.3
- Responsive design framework
- Tableau JavaScript API for embed integration

9.2 Project Deliverables

1. **Tableau Workbook:** Published to Tableau Public with 8 visualizations, 1 dashboard, 1 story
2. **Interactive Website:** Responsive HTML site with embedded visualizations
3. **GitHub Repository:** Complete project code and documentation
4. **Project Report:** Comprehensive documentation (this document)
5. **Video Presentation:** 5-7 minute walkthrough of analysis and insights

9.3 Reproducibility

All analysis is fully reproducible:

- Dataset: Publicly available on Kaggle
- Visualizations: Exact specifications documented
- Code: Open source on GitHub
- Tableau workbook: Published with "Show sheets as tabs" enabled

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10. Conclusion

This comprehensive analysis of product placement strategies reveals significant opportunities for retail optimization. Key takeaways include:

Strategic Insights:

- Clothing products consistently outperform other categories, warranting premium placement
- Front of Store positioning provides measurable sales uplift (5-8%)
- Promotional effectiveness varies significantly by category
- Price premium is sustainable without sales penalty

Methodological Contributions:

- Demonstrated effective use of Tableau for retail analytics
- Developed reusable dashboard framework for ongoing analysis
- Created visual storytelling approach for stakeholder communication

Business Impact:

- Actionable recommendations with quantified expected outcomes
- Framework for data-driven merchandising decisions
- Foundation for future predictive modeling initiatives

The interactive dashboards developed in this project provide retail managers with powerful tools for ongoing analysis and decision-making. By leveraging these insights, retailers can optimize product placement strategies, improve promotional effectiveness, and ultimately drive revenue growth.

References

[1] Kaggle. (2024). Impact of Product Positioning on Sales Dataset. Retrieved from https://www.kaggle.com/datasets/amitvkulkarni/impact-of-product-positioning-on-sales

[2] Tableau Software. (2024). Tableau Desktop User Guide. https://help.tableau.com/current/pro/desktop/en-us/

[3] Few, S. (2012). *Show Me the Numbers: Designing Tables and Graphs to Enlighten* (2nd ed.). Analytics Press.

[4] Kirk, A. (2019). *Data Visualisation: A Handbook for Data Driven Design* (2nd ed.). SAGE Publications.

Appendices

Appendix A: Dataset Sample

Sample records from the dataset:

ID	Category	Position	Price	Comp	Promo	Traffic	Sales
1	Clothing	Aisle	27	25	No	High	1,850
2	Electronics	Front	28	26	Yes	Medium	1,780
3	Food	Endcap	29	27	No	Low	1,690

Table 4: Sample dataset records

Appendix B: Tableau Calculations

Average Sales Volume by Category:
AVG([Sales Volume])

Price Differential:
AVG([Price]) - AVG([Competitor's Price])

Appendix C: Color Scheme

Product Category Colors:

- Clothing: Green (#51cf66)
- Electronics: Orange (#ff922b)
- Food: Red (#ff6b6b)

Promotion Colors:

- Yes: Orange (#ff922b)
- No: Teal (#20c997)

Foot Traffic Colors:

- High: Orange/Yellow (#ffd43b)
- Medium: Pink/Red (#ff6b6b)
- Low: Green (#51cf66)

Appendix D: Project Links

Live Demonstration:

- Website: https://abhinav-kodes.github.io/Strategic-product-placement/
- Tableau Public: https://public.tableau.com/app/profile/abhinav.kodes/viz/ProductPlacement_17735108523740

Source Code:

- GitHub Repository: https://github.com/Abhinav-kodes/Strategic-product-placement

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